



COMPETENCY STANDARD

FOR

ELECTRICAL INSTALLATION AND MAINTENANCE (INDUSTRY)

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

Copyright

The Competency Standards for Electrical Installation & Maintenance (Industry) is a document for developing curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

The document was developed under the **Skills for Employment Investment Program (SEIP)** and subsequently reviewed and updated to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the industry's skills requirements in the Light Engineering sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

Public and private institutes may use the information contained in this competency standard for training activities benefitting Bangladesh.

Other interested parties must obtain permission from the owner of this document for reproduction of information in any manner in whole or in part of this Skills Standard, in English or other languages.

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance,

Probashi Kallyan Bhaban (Level-15 & 16)

71-72 Eskaton Garden, Dhaka-1000

Phone: +8802 55138753-55, Fax: +88 02 55138752

Website: <https://sicip.gov.bd/>

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.

- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a glance:

Generic Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-EIM-01-G	Perform Computations Using Basic Mathematical Formulas	1. Identify calculation requirements in the workplace 2. Identify and select appropriate mathematical methods. 3. Use tool/instrument to perform calculations	10
SICIP-LE-EIM-02-G	Communicate in English in the Workplace	1. Read, understand and prepare documents in English 2. Prepare workplace documents in English 3. Perform conversations in English	10
SICIP-LE-EIM-03-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member 4. Solve problems as a team member	10
Total Hour			30

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-EIM-01-S	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-EIM-02-S	Work with Hand Tools, Power Tools and Electrical Testing Instruments	1. Inspect hand tools and power tools for usability 2. Use hand tools, electrical testing instruments properly and safely 3. Operate power tools properly and safely 4. Clean/maintain hand tools and power tools after use	12
SICIP-LE-EIM-03-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	08
Total Hour			30

OCCUPATION SPECIFIC COMPETENCIES (300 HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-EIM-01-O	Perform Channel Wiring	1. Interpret fundamental principles of electricity and electronics 2. Carry out basic electrical circuit 3. Perform cable joints 4. Understand different types of electrical loads and load calculation 5. Carry out channel wiring 6. Maintain and store electrical/ electronic tools/instruments	80
SICIP-LE-EIM-02-O	Perform Service Connection	1. Interpret drawings and specifications 2. Measure the distance of service line 3. Install cables for service connection 4. Install energy meter 5. Connect energy meter and main switch	20
SICIP-LE-EIM-03-O	Install Earthing and Atmospheric Lightning Protection System	1. Identify type of earthing to be used 2. Identify type of lightning protection system 3. Select and collect tools, equipment and materials 4. Excavate hole for earthing element installation 5. Install earthing components 6. Finish earth pit chamber for pipe earthing method 7. Install lightning protection system 8. Clean and maintain the work area	20
SICIP-LE-EIM-04-O	Perform Motor Connection	1. Identify and select control and protective devices 2. Collect tools, equipment and materials 3. Install control and protective devices 4. Perform motor connection 5. Check and test circuit 6. Clean the workplace	80
SICIP-LE-EIM-05-O	Perform Motor	1. Check the motor physically	80

Code	Unit of Competency	Elements of Competency	Duration (hours)
	Rewinding and Servicing	- and dismantle it to detect the actual fault 2. Select tools and prepare material for winding/rewinding 3. Carry out winding/rewinding of armature for single phase motor 4. Carry out winding/rewinding of stator for three-phase motor 5. Perform connections, carry out pre-assembly tests and assemble machine 6. Carry out final test and record the test result	
SICIP-LE-EIM-06-O	Install and Troubleshoot Solar Electrical System	1. Estimate electrical load for solar system installation 2. Identify tools, equipment and materials 3. Set solar panel 4. Install solar system and accessories 5. Diagnose faults in solar system unit and wiring 6. Repair faults in solar system unit and wiring 7. Clean and store the tools and materials	20
Total Hours			00

COMPETENCY STANDARD: ELECTRICAL INSTALLATION AND MAINTENANCE

A. The Generic Competencies

Unit of Competency: PERFORM COMPUTATIONS USING BASIC MATHEMATICAL FORMULAS	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-EIM-01-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, identifying and selecting appropriate mathematical methods, and using appropriate tools and instruments to perform calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 <u>Calculation requirements</u> are identified 1.2 <u>Workplace information</u> are interpreted
2. Identify and select appropriate mathematical methods	2.1 <u>Appropriate calculation method</u> is identified 2.2 Appropriate calculation method is selected
3. Use tool/instrument to perform calculations	3.1 Appropriate <u>tools and instruments</u> are identified 3.2 Computation is made

Range of variables:

Variable	Range
	May include but not limited to:
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: Length/Width/Height/thickness/diameter 1.4 Weight and density Calculation
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition, Subtraction, Division and Multiplication 3.2 Conversion of measurement 3.3 Percentage and ratio calculation 3.4 Geometric Formulas (Square, Rectangle, Circle, Sphere, etc.)

4. Tools and instruments	4.1 Calculator 4.2 Computer
--------------------------	--------------------------------

Curricular Content Guide

1. Underpinning Knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods, such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas 1.3 Mathematical language, symbols 1.4 Measuring units
2. Underpinning Skills	2.1 Computing basic mathematics 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume, etc. using formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect for rights of peers, sub-ordinates and seniors in the workplace 3.5 Environmental concerns 3.6 Sincerity and honesty to duties 3.7 Eagerness to learn 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Stationeries 4.2 Consumables 4.3 Calculators 4.4 Computers 4.5 Measuring tape

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified calculation requirements from workplace information. 1.2 Selected appropriate method to carry out the calculation requirements. 1.3 Completed calculations using appropriate tools/instruments.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of	3.1 Competency assessment must be done in an

Assessment	assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert
------------	--

Unit of Competency: COMMUNICATE IN ENGLISH IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-EIM-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading, understanding and preparing documents in English, preparing workplace documents in English and performing conversations in English.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read, understand and prepare documents in English	1.1 Workplace documents are read and understood 1.2 Visual information is interpreted 1.3 Resume/CV is understood and prepared
2. Prepare workplace documents in English	2.1 Simple <u>routine workplace documents</u> are prepared using key words, phrases and simple sentences 2.2 <u>Visual aids</u> are prepared and submitted 2.3 Key information is written in the appropriate places in standard forms
3. Perform conversations in English	4.1 Active listening is demonstrated 4.2 Conversation is performed in English with peers, customers and management as per the workplace standard

Range of Variables

Variable	Range
	May Include but not limited to:
1. Routine workplace documents	1.1 Meeting Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional materials 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs
2. Visual aids	2.1 Maps 2.2 Diagrams 2.3 Forms 2.4 Labels 2.5 Graphs 2.6 Charts

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Interpreting workplace documents in English 1.2 Preparing workplace documents in English 1.3 Listening to conversation in English 1.4 Job roles, responsibilities and compliances 1.5 Understanding of visual aids (e.g., graphs, presentations, or charts) and their preparation 1.6 Knowledge of workplace communication etiquette (e.g., formal vs. informal conversations) 1.7 Awareness of workplace standards for professional communication
2. Underpinning Skills	2.1 Reading, understanding and preparing documents in English 2.2 Preparing workplace documents in English 2.3 Performing conversations in English language 2.4 Interpersonal skills to maintain polite and professional communication in workplace settings
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect for rights of peers, sub-ordinates and seniors in the workplace 3.5 Environmental concerns 3.6 Sincerity and honesty to duties 3.7 Eagerness to learn 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Work place Procedure 4.2 Materials relevant to the proposed activity 4.3 All tools, equipment, materials and documentations required 4.4 Relevant specifications or work instructions

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Read, understood and prepared documents in English 1.2 Conversed in English with peers and customers 1.3 Made reports of workplace documents in English
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert
---------------------------------	--

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-EIM-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practice. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range
1. Forms of communication	May Include but not limited to: 1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional materials 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and achieving objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice 2.8 Appreciating and valuing team member's support to other members 2.9 Identifying current and potential problems faced by the

	team 2.10 Identifying solutions to the problems 2.11 Solving problems effectively and evaluating the outcome of the implemented solutions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect for rights of peers, sub-ordinates and seniors in the workplace 3.5 Environmental concerns 3.6 Sincerity and honesty to duties 3.7 Eagerness to learn 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

B. The Sector Specific Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-EIM-01-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> are selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines

2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.3.1 Machine hazards 3.3.2 Materials hazards 3.3.3 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Explosion response 1.3.4 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 PPE types and uses 1.6 Personal hygiene practices 1.7 OHS awareness
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities 4.2 Materials are relevant to the proposed activity 4.3 Workplace (simulated or actual) 4.4 PPEs 4.5 Firefighting equipment 4.6 Emergency response manual 4.7 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures. 1.4 Reported incidents arising from hazards and risks to the authority. 1.5 Implemented plans and procedures to respond emergency 1.6 Applied basic first aid procedure
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Unit of Competency: WORK WITH HAND TOOLS, POWER TOOLS AND ELECTRICAL TESTING INSTRUMENTS	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-EIM-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools, power tools and electrical testing instruments properly. It specifically includes the tasks of inspecting hand tools, power tools and electrical testing instruments for usability, using hand tools and electrical testing instruments properly and safely, operating power tools properly and safely and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools, power tools and electrical testing instruments for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 <u>Hand tools and electrical testing instruments/measuring devices</u> are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools and electrical testing instruments properly and safely	2.1 Appropriate hand tools, electrical testing instruments for the job are used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 <u>Safety precautions</u> is observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing.
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance with manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned. 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 Measuring tools are checked and calibrated.

	4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.
--	--

Range of Variables

Variable	Range
	May include but not limited to:
1. Hand tools and electrical testing instruments/measuring devices	1.1 tools: 1.1.1 Side cutter pliers 1.1.2 Set of screw drivers 1.1.3 Set of combination wrenches 1.1.4 Adjustable wrench 1.1.5 Wire stripper 1.1.6 Electrician's knife 1.1.7 Measuring tape 1.1.8 Electrical Hand Drill 1.2 Electrical testing Instruments: 1.2.1 Multi-meter/AVO meter 1.2.2 Clamp meter 1.2.3 Volt meter 1.2.4 Ammeter Power meter 1.2.1 Megger tester 1.2.2 Soldering iron with stand 1.2.3 Solder sucker 1.2.4 Neon tester 1.3 Mechanical Tools: 1.3.1 Cutting pliers 1.3.2 Nose pliers 1.3.3 Combination pliers 1.3.4 Measuring tape 1.3.5 Tweezers 1.3.6 Scissors 1.3.7 Wire striper 1.3.8 Power Saw 1.3.9 Drill Machine
2. Safety precautions	3.1 Use of appropriate PPEs 3.2 Proper hand, feet and eye coordination 3.3 Working environment 3.4 Safe operating condition for hand tools and power tools 3.5 Awareness of OHS requirements

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of tools, functions and use 1.2 Types of Hand tools and their proper use and
---------------------------	---

	techniques 1.3 Types of electrical testing instruments and their functions 1.4 Types of Power tools, use and safe handling method 1.5 Technical application of tools 1.6 Procedures in the use of hand tools and power tools 1.7 Policies and procedures for occupational health and safety 1.8 Reporting and documentation 1.9 Preventive maintenance 1.10 Methods and techniques 1.11 Quality procedures 1.12 Waste management 1.13 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tools for the job 2.2 Observing safety precautions when using hand tools 2.3 Using electrical testing instruments or devices 2.4 Using power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Checking condition of tools after use 2.6 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.7 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.8 Storing Tools, electrical testing instruments/devices and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers, sub-ordinates and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance with manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories. 1.6 Stored tools and power tools safely in appropriate location.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 08 hrs.	Unit Code: SICIP-LE-EIM-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental

	- impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses, like use of fossil fuel, steel cement, etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Light Engineering sector 1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 1.3 Sources of environmental impacts in Light Engineering 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Light Engineering activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely disposal of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers, sub-ordinates and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Light Engineering sector 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

C. Occupation-specific Competencies

Unit of Competency: PERFORM CHANNEL WIRING	Nominal Duration: 80 hrs	Unit Code: SICIP-LE-EIM-01-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform channel wiring. It specifically includes the tasks of interpreting fundamental principles of electricity and electronics, carrying out basic electrical circuit, performing cables joints, understanding different types of electrical loads and load calculation, carrying out channel wiring, and maintaining and storing electrical/electronic tools/instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret fundamental principles of electricity and electronics	1.1 <u>Fundamental principles</u> of electricity and electronics are described. 1.2 Application of Ohm's law is explained. 1.3 <u>Electrical/Electronic properties/parameters</u> are measured using appropriate measuring tool/instrument or comprehended
2. Carry out basic electrical circuit	2.1 <u>Basic elements</u> for preparing electrical circuits are identified and collected 2.2 Soldering is performed and practiced. 2.3 <u>Electrical/electronic circuit diagram</u> is understood and analyzed. 2.4 <u>Basic electrical circuit</u> wirings are carried out.
3. Perform cable joints	3.1 Different types of <u>cable joints</u> used in industry are explained. 3.2 Characteristics and sizes of the cables are interpreted. 3.3 Cable joints are carried out as workplace standard.
4. Understand different types of electrical loads and load calculation	4.1 <u>Resistive loads</u> are identified and explained 4.2 <u>Inductive loads</u> are identified and explained 4.3 <u>Capacitive loads</u> are identified and explained
5. Carry out channel wiring	5.1 Wiring layout is drawn according to supplied drawing. 5.2 Rowel plug points are located, drilled and inserted as per procedure. 5.3 Bottom parts of the channels are installed and screwed. 5.4 Cables with ECC are laid on the bottom part of the channel 5.5 <u>Boards and different fixtures</u> are collected, fitted and connected as per diagram.

	5.6 The bottom parts of the channels are covered with upper part of the channel
6. Maintain and Store electrical/electronic tools/instruments	6.1 Electrical/electronic tools/instruments are checked for proper operation. 6.2 Electrical/electronic tools/instruments are maintained in accordance to manufacturer's specification. 6.3 Electrical/electronic tools/instruments stored in accordance to workplace procedures/policy.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Fundamental principles	1.1 Ohm's Law 1.2 Kirchhoff's law 1.3 Principles and theory of AC/DC circuits 1.4 Series and parallel circuits 1.5 Law of conductivity 1.6 Law of resistivity 1.7 Semiconductor 1.8 Half wave and full wave rectification
2. Electrical/electronic properties/parameters	2.1 Voltage 2.2 Resistance 2.3 Current 2.4 Impedance 2.5 Capacitance 2.6 Resistance 2.7 Inductance 2.8 Reactance
3. Basic elements	3.1 Switch 3.2 Socket 3.3 Cable 3.4 Resistive load 3.5 Inductive load
4. Electrical/electronic circuit diagram	4.1 Series circuit 4.2 Parallel circuit 4.3 Series-parallel circuit 4.4 Lamp circuits 4.5 Refrigerator circuit 4.6 Window air conditioner circuit 4.7 Deep freezer circuit
5. Basic electrical circuit	5.1 Series circuits 5.2 Parallel circuits 5.3 Series-parallel circuits 5.4 One Lamp controlled from one point 5.5 One Lamp controlled from two points 5.6 Two Lamps controlled from two points 5.7 Two Lamps controlled from three points 5.8 Tube Light Connection

	5.9 Power supply circuit
6. Cable joints	6.1 Straight Through Joint 6.2 Branch Joint 6.3 Tee Joint 6.4 Tap Joint 6.5 Soldered Joint
7. Resistive loads	2.1 Electric bulb 2.2 Heater coil 2.3 Electric Iron 7.1 Rice Cooker
8. Inductive loads	2.4 Induction motor 2.5 Electric fan 2.6 Magnetic Conductor 8.1 Electric choke coil/Ballast
9. Capacitive loads	2.7 Capacitor load bank 9.1 PFI plant
10. Boards and different fixtures	10.1 Boards: 10.1.1 Plastic board 10.1.2 Ebonite boards 10.1.3 Wooden boards 10.2 Different Fixtures: 10.2.1 Switches 10.2.2 Sockets 10.2.3 Fan regulator 10.2.4 Fuse 10.2.5 Ceiling rose 10.2.6 Indicating lamp 10.2.7 Lamp Holders 10.2.8 LED Lights 10.2.9 Tube Light 10.2.10 Circuit Breakers (CB) 10.2.11 MCCB 10.2.12 ELCB

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental principles of electricity and electronics 1.2 Definition of AC and DC current and their uses in circuits 1.3 Types of DC circuits and their application 1.4 AC circuits and their application 1.5 Differential electrical/electronic circuit diagrams
---------------------------	---

	applied in refrigeration and air conditioning 1.6 Circuit wiring, installation and maintenance 1.7 Cable joints and different types of joints 1.8 Resistive, inductive and capacitive loads and their uses in different circuits 1.9 Basic wiring circuits and their application 1.10 Electrical measurement and testing methods and techniques 1.11 Safety precautions when working with electrical circuits 1.12 Channel wiring and testing methods
2. Underpinning Skills	2.1 Carrying out basic electrical circuit diagramming and wiring 2.2 Describing relationships of the different types of electrical properties 2.3 Measuring electrical parameters using appropriate measuring tool/instrument 2.4 Using electrical/electronic measuring tools and testing instruments safely/properly 2.5 Preparing basic electrical circuits 2.6 Performing channel wiring and testing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Various kinds of electrical/electronic components, tools and equipment 4.3 Workplace rules and regulation policy manual 4.4 Pens 4.5 Papers 4.6 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Carried out basic electrical circuit diagramming and wiring 1.2 Performed different types of cable joints 1.3 Measured electrical parameters using appropriate
-----------------------------------	---

	measuring tool/instrument 1.4 Used resistive, inductive and capacitive loads in different electrical/electronic circuits and performed testing with instruments safely/properly 1.5 Performed channel wiring and testing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SERVICE CONNECTION	Nominal Duration: 20 hrs	Unit Code: SICIP-LE-EIM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform service connection. It specifically includes the tasks of interpreting drawings and specifications, measuring the distance of service line, installing cables for service connection, installing energy meter and connecting energy meter and main switch.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret drawings and specifications	1.1 <u>Drawings</u> are collected and interpreted. 1.2 <u>Signs and symbols</u> are identified. 1.3 Terms and abbreviations are identified. 1.4 Specifications are interpreted.
2. Measure the distance of service line	2.1 PPEs are collected and used. 2.2 Distance between distribution pole and meter are checked and measured. 2.3 Distance between main switch and meter are checked and measured.
3. Install cables for service connection	3.1 Size of cables are selected as per load. 3.2 Collected cables are cut and set. 3.3 Cables are held on and clamped properly with distribution pole. 3.4 Cables are joined and connected with the pole and energy meter. 3.5 <u>Materials</u> for the connection are selected
4. Install energy meter	4.1 Energy meter is collected and set on the board. 4.2 Energy meter is connected with service line.
5. Connect energy meter and main switch	5.1 Cables are measured and sized. 5.2 Cables are laid into the conduit. 5.3 Connection between energy meter and main switches is performed.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Drawings	1.1 Sketch 1.2 Electrical drawings
	2.1 Drawing symbol 2.2 Connection symbols 2.3 Load symbols

2. Signs and symbols	2.4 Socket symbols 2.5 Main switch symbols 2.6 Supply symbols 2.7 Danger symbols 2.8 Switch board symbols 2.9 Conduit board 2.10 Circuit Breaker symbol 2.11 Protective device symbol
3. Materials	3.1 GI wire 3.2 Connector 3.3 Distribution board 3.4 Energy Meter 3.5 Main switch 3.6 Cables 3.7 Insulator 3.8 Clamps 3.9 Tie 3.10 Hook

Curricular Content Guide

1. Underpinning Knowledge	Trainee will acquire knowledge of: 1.1 Layout drawing, signs and symbols 1.2 Measurements related to service connections 1.3 Materials to be used 1.4 Service cables and energy meter
2. Underpinning Skills	2.1 Collecting & preparing materials 2.2 Measuring distance 2.3 Setting service cables 2.4 Connecting service cables, energy meter and main switch
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace

4. Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools and equipment appropriate to joint and connection process 4.3 Availability of materials
--------------------------	---

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Installed service cable, energy meter and main switch as per drawing 1.2 Connected energy meter and main switch as per drawing
2. Methods of Assessment	2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL EARTHING AND ATMOSPHERIC LIGHTNING PROTECTION SYSTEM	Nominal Duration: 20 hrs	Unit Code: SICIP-LE-EIM-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to install earthing and atmospheric lightning protection system. It specifically includes the tasks of identifying type of earthing to be used, identifying type of lightning protection system, selecting and collecting tools, equipment and materials, excavating hole for earthing element installation, installing earthing components, finishing earth pit chamber for pipe earthing method, installing lightning protection system, and cleaning and maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify type of earthing to be used	1.1 Types and <u>methods of earthing</u> are identified in accordance with electrical plan/design 1.2 Types and sizes of <u>earthing materials</u> are identified in accordance with electrical plan/design
2. Identify type of lightning protection system	1.3 <u>Types of lightning protection system(LPS)</u> is identified in accordance with electrical plan/design 1.4 Types and sizes of <u>lightning protection system materials</u> are identified in accordance with electrical plan/design
3. Select and collect tools, equipment and materials	1.5 <u>Tools and equipment</u> are selected and collected 1.6 Tools and equipment are checked for usability 1.7 Earthing materials are collected and checked for conformance in accordance to specification 1.8 Lightning protection materials are collected and checked for conformance in accordance to specification
4. Excavate hole for earthing element installation	1.9 <u>PPE</u> is collected and used in accordance with OHS requirements 1.10 Hole is dug following the safety requirements 1.11 Hole is shaped and sized in accordance with electrical plan/design and specification
5. Install earthing components	1.12 Earthing element is fitted in the bottom of the excavated hole following standard earthing procedure 1.13 Earth lead is connected to the earth element tightly

	and brought up the meter board through the conduit. 1.14 Powdered charcoal and salt are laid around the earthing element in accordance with workplace procedure. 1.15 A GI pipe of proper size and length is fitted from top of the earth element to the bottom of the earth pit chamber. 1.16 Rest of the excavated hole is filled with earth.
6. Finish earth pit chamber for pipe earthing	1.17 Earth pit chamber is constructed with brick chips, cements sand and water mixture in accordance with standard/specification. 1.18 Pit chamber cover is made with GI sheet in accordance with electrical plan/design. 1.19 Pit cover is fitted/installed on the pit chamber. 1.20 Check earth resistance in accordance with electrical plan/specification
7. Install lightning protection system	1.21 Lightning rod is installed at specified location. 1.22 Earth down conductor is connected as per diagram. 1.23 Performance of lightning protection system (LPS) is tested as per standard.
8. Clean and maintain the work area	1.24 Electrical tools/instruments are cleaned and checked for operability 1.25 Work area is cleaned and waste materials are disposed in accordance with workplace requirements

Range of Variables

Variable	Range (Includes but not limited to:)
1. Methods of earthing	1.1 Pipe earthing 1.2 Rod earthing 1.3 Plate earthing 1.4 Waterman earthing 1.5 Strip or wire earthing
2. Earthing materials	2.1 Continuity conductor/cable 2.2 Earthing lead 2.3 Earth electrode / plate 2.4 Connector 2.5 G.I. pipe / conduit 2.7 Bolts and nuts 2.8 Powdered charcoal 2.9 Salt

3. Tools and equipment	3.1 Tools: 3.1.1 Measuring tape (30m) 3.1.2 Tri- square 3.1.3 Pocket tape (3m) 3.1.4 Claw hammer / crow bar 3.1.5 Wire stripper 3.1.6 Adjustable wrench 3.1.7 Bolt cutters 3.1.8 C-clamp 3.1.9 Chisels: (wooden, cold) 3.1.10 Drill bits 3.1.11 Files: (flat, round, half round) 3.1.12 Hand hacksaw 3.1.13 Hammers: (ball peen, claw) 3.1.14 Pliers (combination pliers, cutting pliers, diagonal cutting pliers, long nose pliers) 3.1.15 Punches 3.1.16 Screwdrivers (star, negative, positive) 3.1.17 Electrician knife 3.2 Equipment: 3.2.1 Electric drill machine 3.2.2 Soldering iron 3.2.3 Megger tester 3.2.4 Multi meter/AVO meter 3.2.5 Earth Tester
4. PPE	4.1 Safety shoes 4.2 Safety gloves 4.3 Safety helmet 4.4 Uniform
5. Types of lightning protection system (LPS)	5.1 Conventional LPS 5.2 Non-conventional LPS
6. Lightning protection system materials	6.1 Lightning rod (spike arrester) 6.2 Earth down conductor (arrester) 6.3 Check terminal 6.4 Earth lead

Curricular Content Guide

1. Underpinning Knowledge	Trainee will acquire knowledge of: 1.1 Types of earthing 1.2 Materials required for earthing
---------------------------	--

	1.3 Procedure of earthing 1.4 Testing procedure of earthing 1.5 Types of LPS 1.6 Materials required for LPS 1.7 Installation procedure of LPS 1.8 Performance testing procedure of LPS
2. Underpinning Skills	1.9 Collecting earthing and LPS materials and checking for conformance in accordance with specification 1.10 Excavating a hole for the earthing element 1.11 Fitting earthing element in the bottom of the excavated hole following standard earthing procedure 1.12 Connecting earth led to the earth element tightly and brought up the meter board through the conduit 1.13 Installing lightning rod at specified location 1.14 Connecting earth down conductor as per diagram 1.15 Testing the performance of lightning protection system (LPS) as per standard 1.16 Checking earth loop resistance using appropriate test instrument
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools and equipment required for the job 4.3 Materials required 4.4 Work books 4.5 Diagram and instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS regulations for the required job 1.2 Tested earth resistance as per standard procedure 1.3 Tested the performance of LPS as per standard
-----------------------------------	---

	procedure
2. Methods of Assessment	Competency should be assessed by: 2.5 Written Test 2.6 Practical Demonstration 2.7 Oral questioning 2.8 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	Nominal Duration:	Unit Code:
----------------------------	--------------------------	-------------------

PERFORM MOTOR CONNECTION	80 hrs	SICIP-LE-EIM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform motor connection. It specially includes the tasks of identifying and selecting control and protective devices, collecting tools, equipment and materials, installing control and protective devices, performing motor connection, checking and testing circuit, and cleaning the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and select control and protective devices	1.1 <u>Manuals</u> and documents of controlling and protective devices are collected. 1.2 <u>Drawings</u> and <u>symbols</u> of controlling and protective devices are sorted. 1.1 Types of <u>controlling</u> and <u>protective devices</u> are listed.
2. Collect tools, equipment and materials	2.1 Tools, equipment and materials are identified and collected. 2.2 Tools, Equipment and Materials are checked for usability, 2.3 PPE is collected and used.
3. Install control and protective devices	3.1 Control and <u>protective</u> devices are selected and collected according to the need of the operations. 3.2 Control and protective devices are installed according to the layout plan. 3.3 Control and protective devices are set and connected to the motor.
4. Perform motor connection	4.1 Direct on-line starter is collected and its diagram is interpreted. 4.2 Direct on-line starter is connected with the motor. 4.3 Star-delta starter is collected and its diagram interpreted. 4.4 Star-delta starter is connected with the motor. 4.5 Reverse-Forward starter is collected and its diagram interpreted. 4.6 Reverse-Forward starter is connected with the motor.
5. Check and test circuit	5.1 All the connections of each starter are checked and justified. 5.2 Connection between motor and starter is checked and tested.
6. Clean the workplace	6.1 Cleaning tools and equipment are selected & collected. 6.2 Cleaning tools and equipment are prepared for

	cleaning. 6.3 Waste materials are disposed. 6.4 Cleaning is completed.
--	--

Range of Variables

Variable	Range (Includes but not limited to:)
1. Manuals	1.1 Manufacturer's specification manual 1.2 Repair manual 1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Manual of instruction
2. Drawings	2.1 Technical drawing 2.2 Sketch 2.3 Blue print 2.4 Electrical drawings 2.5 Connection diagram
3. Symbols	3.1 Drawing symbol 3.2 Connection symbol 3.3 Load symbol 3.4 Socket symbol 3.5 Main switch symbol 3.6 Supply symbol 3.7 Danger symbol 3.8 Switch board symbol 3.9 Conduit symbol 3.10 Starter symbol 3.11 Protective device symbol 3.12 Motor symbol
4. Controlling Devices	4.1 Push Switches 4.2 Main switches 4.3 Direct on-line starter 4.4 Change over switches 4.5 Timer 4.6 Relay 4.7 Magnetic contactor
5. Protective devices	5.1 Circuit breaker 5.2 Over load Relays 5.3 MCCB 5.4 ELCB

Curricular Content Guide

1. Underpinning Knowledge	Trainee will acquire knowledge of: 1.1 Diagram of motor connection 1.2 Diagram of protective and controlling devices 1.3 Classification of protective and controlling devices 1.4 Drawings and manuals 1.5 Technique of motor connection
2. Underpinning Skills	2.1 Collecting manuals and diagrams required for motor connection 2.2 Collecting and using PPE 2.3 Collecting and using tools and equipment as per job requirements 2.4 Collecting and using materials as per job requirements 2.5 Installing service cables, main switch, Starter and motor 2.6 Performing motor connection
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Materials, tools and equipment needed for the activity 4.3 Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.17 Installed protective and controlling devices 1.18 Connected motor
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
--	--

Unit of Competency: PERFORM MOTOR REWINDING AND SERVICING	Nominal Duration: 80 hrs	Unit Code: SICIP-LE-EIM-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform motor rewinding and servicing. It specially includes the tasks of checking the motor physically and dismantling it to detect the actual fault, selecting tools and preparing material for winding/rewinding, carrying out winding/rewinding of armature for single phase motor, carrying out winding/rewinding of stator for three-phase motor, performing connections, carrying out pre-assembly tests and assembling the machine, and carrying out final test and recording the test result.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Check the motor physically and dismantle it to detect the actual fault	1.1 The motor is checked physically and dismantled to detect the actual fault. 1.2 Visual and manual inspection is carried out to detect the mechanical damage/defects. 1.3 Burnt winding is checked by smelling. 1.4 The machine is dismantled as per standard procedure and manufacturer's instructions. 1.5 Winding data is collected and winding diagram is prepared. 1.6 Winding is checked with specified <u>measuring instruments</u> to detect the fault. 1.7 Associated accessories' faults are checked as per manufacturer's data and noted down. 1.8 Bearing, carbon brushes, rockers, slip rings are checked visually and tested by specified instrument. 1.9 Fault is detected and noted down for proper repair.
2. Select tools and prepare material for winding/rewinding	2.1 Technical information and winding data are collected and winding diagram is prepared as per winding data 2.2 Specified <u>tools</u> , <u>materials</u> and <u>equipment</u> are selected for winding work. 2.3 Winding wire of required gauge, insulation and binding material are prepared. 2.4 Formers are prepared according to winding data. 2.5 Coil is formed on former as per collect winding data.
3. Carry out winding/rewinding of armature for single	3.1 Armature are cleaned by using specified tools. 3.2 Insulation material is inserted into slots 3.3 Formed coil is inserted into slots as per standard

phase motor	3.4 Rotor is cleaned and checked 3.5 Servicing and repaired works is carried out as necessary 3.6 Formed coil is insulated, varnished, heated and inserted into slot
4. Carry out winding/rewinding of stator for three-phase motor	4.1 Stator are cleaned by using specified tools. 4.2 Insulation material is inserted into slots 4.3 Formed coil is inserted into slots as per standard 4.4 Rotor is cleaned and checked 4.5 Servicing and repaired works is carried out as necessary 4.6 Formed coil is insulated, varnished, heated and inserted into slot
5. Perform connections, carry out pre-assembly tests and assemble the machine	5.1 Winding resistance and insulation level is checked by specified test instrument. 5.2 Coil ends are terminated and insulated. 5.3 Winding continuity is checked. 5.4 Dry running is carried out, performance is checked and remedial action has taken, if necessary. 5.5 Insulating varnish is applied to winding and baked it up to recommended temperature in baking oven. 5.6 Assembly of stator, rotor and other parts are assembled.
6. Carry out final test and record the test result	6.1 Rotor free movement is checked. 6.2 No load/load/locked rotor tests are performed. 6.3 Rotor static and dynamic balancing is carried out. 6.4 Test results are documented in the relevant recorded sheet. 6.5 Test reports are prepared. 6.6 Machine performance is demonstrated. 6.7 Tools, equipment and materials are cleaned and stored as per workplace standard.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Equipment	1.1 Electric oven, manual rewinding machine 1.2 Power driven rewinding machine 1.3 Winding head cutter 1.4 Winding puller 1.5 Electric hand drill machine with bits 1.6 AC & DC power supply unit 1.7 Single phase AC motor

	1.8DC motor 1.9Three phase AC motor, universal motor
2. Measuring Instruments	1.10 Ammeters 1.11 Voltmeters 1.12 Ohmmeter 1.13 Wattmeter 1.14 Megger 1.15 Clamp-on-AVO meter 1.16 AC & DC power supply unit
3. Tools	Toolbox containing a set of tools. (The following tools may include to make a set but not limited to): 1.17 Flat/Philips headed screw drivers 1.18 Bradawl 1.19 Electrician knife 1.20 Diagonal cutting pliers 1.21 Combination pliers 1.22 Round nose pliers 1.23 Stripping pliers 1.24 Adjustable wrenches 1.25 Hacksaw 1.26 Hammer 1.27 Files 1.28 Measuring tape 1.29 Steel ruler 1.30 Soldering iron 1.31 Neon line tester 1.32 Spirit level 1.33 Continuity tester 1.34 Socket wrenches sets 1.35 Bearing puller 1.36 Crimping pliers 1.37 Saw 1.38 Divider 1.39 Micrometer 1.40 Bolt cutter 1.41 Pipe cutter 1.42 Manual hand drill 1.43 Percussion drill set.
4. Materials	1.44 Super enamel wires

	1.45 Insulating plastic film 1.46 PVC tape 1.47 Laminates 1.48 Insulating paper 1.49 Ceramic fiber flexible cables 1.50 Slot insulation paper 1.51 Binding thread 1.52 Varnish 1.53 Thinner 1.54 Grease 1.55 Bearings 1.56 Solder 1.57 Resin 1.58 Winding string
--	---

Curricular Content Guide

1. Underpinning Knowledge	Trainee will acquire knowledge of: 1.1 Estimation of winding. 1.2 Types of winding. 1.3 Process of preparation of insulation paper, coils and wedges. 1.4 Setting the insulation paper, coils and wedges. 1.5 Internal position of the coil groups. 1.6 Process of connection of coils (winding) in groups. 1.7 Separation of groups of coils. 1.8 Process of connection of the winding terminals to the terminal post. 1.9 Process of preparation and application of varnish. 1.1 Measurement of insulation resistance
2. Underpinning Skills	1.10 Using the proper tools 1.11 Dis-assembling the motor 1.12 Tracing out the internal connection of the winding 1.13 Preparation of the insulation paper coils and wedges 1.14 Setting the insulation paper, coils and wedges in the slots 1.15 Testing the connection of the coils with meter 1.16 Assembling the motor for test run 1.17 Preparation of the motor for varnishing 1.18 Assembling the motor for use

3. Underpinning Attitudes	1.19 Commitment to occupational health and safety 1.20 Promptness in carrying out activities 1.21 Sincerity and honesty to duties 1.22 Environmental concerns 1.23 Eagerness to learn 1.24 Tidiness and timeliness 1.25 Respect for rights of peers and seniors in workplace 1.26 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided; 1.27 Workplace (actual or simulated) 1.28 Materials relevant to the proposed activity 1.29 Measuring instrument 4.1 Relevant drawings, manuals, codes, standards and reference materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified winding faults 1.2 Formed coil for rewinding as per job requirements 1.3 Inserted coil in the slot and completed connections 1.4 Tested motor before establishing electrical connection
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL AND TROUBLESHOOT SOLAR ELECTRICAL SYSTEM	Nominal Duration: 20 hrs	Unit Code: SICIP-LE-EIM-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to install and troubleshoot solar electrical system. It specially includes the tasks of estimating electrical load for solar system installation, identifying tools, equipment and materials, setting solar panel, installing solar system and accessories, diagnosing faults in solar system unit and wiring, repairing faults in solar system unit and wiring, and cleaning and storing the tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Estimate electrical load for solar system installation	1.1 Required <u>electrical loads</u> are estimated. 1.2 Layout drawing of selected work plan is prepared. 1.3 Capacity of panel, battery, inverter, charge controller and <u>other accessories</u> are selected as per load. 1.4 Following the layout plan, required quantity and size of cable, wire and other installation materials are estimated. 1.5 Information on cost of <u>equipment, accessories and materials</u> are collected. 1.6 Cost of equipment, accessories and materials are estimated. 1.7 Installation charges are estimated.
2. Identify tools, equipment and materials	2.1 <u>Tools</u> are selected and collected. 2.2 <u>Installation materials</u> and <u>solar electrical system components and accessories</u> are collected. 2.3 Battery is collected and tested.
3. Set solar panel	3.1 Personal Protective Equipment (PPE) is used while working. 3.2 Special rope, safety belts and ladder are used for working on the roof. 3.3 Frames are constructed as per panel size. 3.4 Appropriate place with maximum sunlight exposure for panel setting is located. 3.5 Setting of panels within frame is demonstrated between 23 to 30 degrees.
4. Install solar system and accessories	4.1 Solar system and accessories are installed as per layout plan.

	4.2 Channel or conduit wiring is performed as per layout diagram. 4.3 Switches and sockets on the board are fixed. 4.4 Connections with all related components are performed. 4.5 Testing of solar electrical system for operation is performed.
5. Diagnose faults in solar system unit and wiring	5.1 <u>Physical faults</u> in the inverter, charger, charge controller, panel, battery and wiring system are checked visually. 5.2 <u>Operational faults</u> in the inverter and charge controller are checked by <u>testing instruments</u>. 5.3 Panel is tested for appropriate functioning. 5.4 Battery is checked by meter for appropriate voltage. 5.5 Electrolyte of battery is checked by hydrometer. 5.6 <u>Electrical connections</u> are checked throughout the wiring. 5.7 Charge controller and inverter are tested.
6. Repair faults in solar system unit and wiring	6.1 Burnt components are replaced. 6.2 Inactive and faulty components are replaced. 6.3 Battery water is added, if required. 6.4 Loose Connections are repaired throughout the wiring. 6.5 Charge controller is repaired.
7. Clean and store the tools and materials	7.1 Tools and equipment are cleaned. 7.2 Tools, measuring instruments and access materials are stored as per workplace standards.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Electrical Load	1.1 Light 1.2 Ceiling fan 1.3 Television 1.4 Refrigerator 1.5 Water pump 1.6 Computer
2. Other accessories	1.7 Light fixtures 1.8 Switch board 1.9 Switch 1.10 Sockets

	1.11 MCB 1.12 Cables and wires
3. Equipment, accessories and materials	1.13 Solar panel 1.14 Charge controller 1.15 Battery 1.16 Inverter 1.17 Switch and sockets 1.18 Cables/wires 1.19 Conduit 1.20 Fixing materials
4. Tools	1.21 Screw drivers 1.22 Diagonal cutting pliers 1.23 Long nose pliers 1.24 Combination pliers 1.25 Adjustable wrenches 1.26 Socket wrench set 1.27 Hammer 1.28 Electric hand drill with bits 1.29 Sprit level 1.30 Measuring tape 1.31 Neon tester 1.32 Battery tester
5. Installation materials	1.33 Cables 1.34 Channels 1.35 Screws 1.36 Rawl plugs 1.37 Clips 1.38 Nails 1.39 Plastic board 1.40 Conduits 1.41 Plastic connectors 1.42 Cable ties
6. Solar electrical system components	1.43 Panel 1.44 Inverter 1.45 Charge controller 1.46 Battery
7. Solar electrical system accessories	1.47 Light fixtures 1.48 Switches 1.49 Sockets 1.50 Junction boxes

8. Physical faults	1.51 Burnt components by high temperature 1.52 Damaged by insect 1.53 Disconnection due to vibration 1.54 Loose connection
9. Operational faults	1.55 Components are inactive by aging 1.56 Components are inactive by transient effects 1.57 Components are inactive due to manufacturing defects 1.58 Components are inactive due to overload
10. Testing instruments	1.59 Volt meter (Analogue/Digital) 1.60 Ammeter (Analogue/Digital) 1.61 Multi meter (Analogue/Digital) 1.62 Hydro meter
11. Electrical connections	1.1 Terminal connections of switches, sockets and light fixtures 1.2 Terminal connection of panel 1.3 Terminal connection of charge controller 1.4 Terminal connection of inverter 1.5 Terminal connection of battery

Curricular Content Guide

1. Underpinning knowledge	Trainee will acquire knowledge of: 1.1 Electrical layout for solar system 1.2 Electrical load for solar system 1.3 Selection of appropriate panel, battery, inverter, charge controller and other accessories 1.4 Techniques of connecting solar home system 1.5 Techniques of testing solar home system 1.1 Different faults in solar home system unit and wiring
2. Underpinning Skills	1.6 Interpreting electrical load 1.7 Preparing layout drawing for selected work 1.8 Constructing frame as per panel size 1.9 Installing solar system and accessories as per layout plan 1.10 Performing solar system testing for operation 1.11 Testing charge controller and inverter 1.12 Replacing burn component 1.13 Repairing loose connection through wiring
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 1.30 Workplace (actual or simulated) 1.31 Materials relevant to the proposed activity 1.32 Measuring instrument 1.33 Relevant drawings, manuals, codes, standards and reference materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Prepared layout drawing for selected work. 1.2 Constructed frame as per panel size. 1.3 Installed solar electrical system and accessories as per layout plan. 1.4 Performed solar home system testing for operation. 1.5 Tested charge controller and inverter. 1.6 Replaced burnt component.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Workshop/Lab Facility Standard

Course Name:	Electrical Installation & Maintenance (Industry)
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

- Classroom – 350 sft
- Workshop/ lab – 1000 sft

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop with internet facility	1
2.	Multimedia projector & screen	1
3.	Wiring Board	20
4.	Motor Coil Rewinding machine	5
5.	Main Distribution Board with busbar, CB and other accessories (Training model)	5
6.	Sub Distribution Board with busbar, CB and other accessories (Training model)	5
7.	DOL Starter panel	4
8.	Star Delta Starter panel	2
9.	Reverse Forward Starter panel	2
10.	Magnetic contactor (coil voltage 220V and 400V)	20
11.	Relay	10
12.	Timer	10
13.	Pilot lamp & Push switch	20
14.	Push switch	30
15.	Generator (Minimum 2 KW)	1
16.	Solar power system set [Solar panel (minimum 40W), battery, charge controller, inverter and load]	2
17.	Electric drill machine	5
18.	Electric grinding machine	2
19.	Soldering Stations	5
20.	Energy Meter (1 Phase)	10
21.	Energy Meter (3 Phase)	5
22.	Multi meter/AVO meter	10
23.	Earth tester	2
24.	Digital weight machine	1
25.	Insulation tester	2
26.	Tachometer	2
27.	Clamp Meter	2
28.	Phase sequence meter	1
29.	Single phase induction motor	4
30.	Three phase induction motor (minimum 3hp)	4

31.	Ceiling fan	10
32.	Table fan	4
33.	Bench Vice	5

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Electrical Installation and Maintenance (Industry), the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.